Forging the Future: Market Intelligence and Growth Strategy for High-Performance Metals Through 2035

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1. Introduction

For centuries, metals have been among the essential materials on which human societies are built, enabling the construction of cities, transport systems, power networks, machinery and technologies that support modern economies. Their uses have expanded with each wave of industrial development, and today they remain central to established industries as well as emerging technologies. This enduring role is reflected in the sector's economic scale: according to McKinsey & Company's Global Materials Perspective, the global metals and mining industry generated approximately US$3.0 trillion in revenue in 2024, after a decline of 6 percent, and grew by only about 1 percent in 2025 (McKinsey & Company, 2025, 2026).

Within this broad market, high-performance alloys serve applications where ordinary materials may not meet the required operating conditions. These engineered materials combine metals such as nickel, cobalt, titanium, iron and chromium to provide properties such as strength at high temperatures, corrosion resistance, oxidation resistance or reliable performance under pressure. Nickel- and cobalt-based superalloys, titanium alloys and corrosion-resistant specialty steels are examples. Demand comes from aerospace and defence, power generation, chemical processing, oil and gas, medical technology and other industries with demanding technical requirements. The production and availability of metals including nickel, cobalt, chromium, molybdenum, titanium, tungsten, zirconium, copper, aluminium and manganese therefore provide important context for understanding this specialist market.

The outlook for high-performance alloys to 2035 is positive, although growth is likely to differ across alloy types and applications. One market estimate places the segment at US$13.6 billion in 2025 and projects it to reach US$19.0 billion by 2034 (The Insight Partners, 2025), although other providers start as low as US$10.2 billion because they draw the boundary of the segment differently (IMARC Group, 2025), while broader industry analysis points to continued demand from aerospace, defence, energy and other industrial uses. This growth will depend on the pace of investment and technology adoption, as well as the cost and availability of alloying metals. The strongest prospects are likely to be in applications where a particular combination of material properties, reliability and qualification requirements creates lasting value for customers.

2. Report Objective and Scope

This project studies selected metals that are essential to high performance alloys and follows them from the mine to the industrial applications that depend on them, in order to assess how their markets may develop through 2035. It asks one central question: how do the concentration of metal supply, the conversion of metals into qualified product forms and the needs of seven industries determine where the most attractive opportunities in high

performance materials will lie through 2035? Four subsidiary questions structure the answer, and each maps onto one part of the report. The first asks where the ten metals are mined and refined, how exposed that supply is to concentration and export policy, and in which qualified forms the metals reach industry. The second asks which industries and regions drive demand for these forms today and how large the measurable market is. The third asks how large demand could become by 2035, how certain that estimate is and which emerging applications could add to it, and the fourth asks where supply security, growth and entry barriers combine into the most attractive opportunities.

The study covers ten metals: nickel, cobalt, chromium, molybdenum, titanium, tungsten, zirconium, copper, aluminium, and manganese. It examines seven industries: automotive, oil and gas, energy and environmental technology, aerospace, chemical processing, electronics and electrical engineering, and medical technology. The study uses the latest official data available in November 2026. It covers the global market and compares Europe, North America, and Asia. The outlook quantifies established uses and evaluates emerging uses separately because official statistics do not yet capture them.

The report argues that much of the value in this market comes from turning metals into qualified products. Nickel is in surplus. Export policies restrict supplies of cobalt, tungsten, and molybdenum. Countries that make qualified bar, strip, and tube earn trade surpluses on those products that mining countries do not.

The most attractive opportunities supported by measured evidence include remelt stock and combustor sheet for gas turbines and triple-melt bar and billet for aircraft engines in the United States and Europe. Chapter 8 also tests the ranking of precision FeNi foil for East Asian mask makers. The tier 1 model summary here gives 2035 median (P50) demand indices of 133 for energy technology, 134 for aerospace, and 102 for automotive, with 2025 set to 100 (Product Strategy 2035 Team, 2026a). Emerging applications appear in a separate register with conditions that would justify moving forward.

The findings suggest that producers should stop investing in parts used only in diesel engines, new European base chemical plants, electrolyzer capacity without funded orders, and captive zirconium for nuclear cladding. Chapter 3 explains the methods. Chapters 4 to 7 examine supply, industry demand, trade, and growth. Chapter 8 ranks the opportunities, and chapter 9 summarizes the findings.

3. Introduction to the Analytical Framework and Methodology

The study follows seven steps in the metal value chain: material, product form, application, industry, geography, market size, and growth. Table 1 shows how the chapters cover these steps. The study compares figures from the same stage. For example, it separates contained

metal from gross weight and actual output from production capacity. Mine output, alloy melt, mill products, and industry revenue are reported separately.

The main sources are structured data from the USGS, UN Comtrade, Eurostat PRODCOM, the World Bank, OECD, UNIDO, and the IEA. The study also uses industry associations, official outlooks, and profiles of 61 companies. Commercial market reports serve as a cross-check. Where a second source exists, the study checks the figures that support its main findings. The source register flags figures that come from a single source and records each value's unit, stage, data type, source, and formula.

Table 1: The metal value chain and its evidence

Link	Guiding question	Main evidence	Chapter
Material	Where is supply, and how concentrated is it?	USGS, IEA, World Bank prices	4
Product form	In which qualified forms does the metal reach industry?	PRODCOM, producer filings	4
Application	Which components need the material, and why?	Standards, associations, filings	4 and 5
Industry	Which industries buy, and how fast do they grow?	Airbus, Boeing, IEA, OICA, Cefic, SEMI	5
Geography	In which regions are the forms bought and made?	IEA regions, OECD, UN Comtrade	5 and 6
Market size	How large is measurable demand today?	Apparent consumption, trade values	5 and 6
Growth	How large could demand be in 2035?	Demand model, emerging options register	7

Tier 1 estimates demand for established applications. It multiplies industry activity by the metal content per unit and by the share of units that use a high-performance alloy. Examples of activity measures include engine deliveries, gigawatts of gas turbines, and joint replacements. The model starts with measured 2025 values, including cobalt use in aviation.

To check the model, the team fitted trends to data through 2019 and compared the projections with actual outcomes from 2020 to 2025. The team also checked earlier official outlooks where available. A Monte Carlo simulation estimates the 10th, 50th, and 90th percentiles of 2035 demand. A separate supply analysis tests how export quotas and licensing changes could affect the results.

Tier 2 records emerging applications that current data cannot confirm. Examples include hydrogen-fueled turbines, supercritical carbon dioxide recuperators, molten salt and small modular reactors, fusion magnet substrate tape, and additively manufactured engine parts. The

team identified candidates from patents, pilot plants, standards under development, and announced projects. For each candidate, it assessed technology readiness, strength of evidence, required alloys and product forms, and the earliest date it could reach commercial volume.

The register identifies an event that could move each candidate into tier 1, such as a final investment decision or a first serial order. Chapter 8 uses the same eight criteria to score both tiers and identifies the evidence behind each score. Producers can use the results to decide which opportunities to monitor and which have enough evidence to consider funding.

4. Metal Markets: Materials, Product Forms and Applications

Table 2 shows the commercial forms of the ten metals covered in this chapter and the main uses reported for each.

Table 2: Commercial forms and main uses by metal

Metal	Commercial forms	Main use and company example	Share
Nickel	Bar Billet Sheet Plate Tube Battery salts	Stainless steel (Alfa Laval T10 316L plate) Batteries (LG Energy Solution M50L cell) Nickel alloys (Haynes C-276 reactor plate)	66% 16% 6%
Cobalt	Remelt stock Bar Powder Cobalt salts	Batteries (LG Energy Solution M50L cell) Superalloys (Haynes 188 combustor sheet)	≈75% ≈7%
Chromium	Ferrochrome Chromium metal Sheet Plate	Stainless steel (Franke MYX 110-55 sink)	>80%
Molybdenum	Ferromolybdenum Plate Sheet Tube	Chemicals & petrochemicals (Alfa Laval T10 C-276 plate) Oil & gas (Alleima SAF 2507 umbilical tube)	17% 14%
Titanium	Sponge Plate Bar Tube Forgings	Aerospace (Airbus A350 landing gear)	>50%*

Metal	Commercial forms	Main use and company example	Share
Tungsten	Carbide powder Inserts Tungsten metal	Carbide products (Sandvik CNMG 4425 insert) Steels & superalloys (Haynes 188 combustor sheet)	65% 14%
Zirconium	Zircon concentrate Sponge Plate Tube	Ceramics (Iluka Premium zircon for tile) Acid service (Alleima Zr 702 exchanger tube) Fuel cladding (Framatome M5 fuel rod cladding)	>50% n.r. n.r.
Copper	Wire Tube Sheet Strip Bar	Construction (Southwire SIMpull THHN wire) Consumer, cooling & electronics (Wieland grooved tube) Power & telecom (Prysmian 12/20 kV Cu cable)	26% 23% 17%
Aluminium	Sheet Plate Extrusions Castings Conductor rod	Construction (WICONA WICLINE 75 Eco frame) Transport (Novelis s118 hood inner sheet) Electrical (Southwire Drake ACSR conductor)	≈25% ≈23% ≈12%
Manganese	Ferromanganese Silicomanganese Electrolytic metal	Steelmaking (voestalpine MN13 rail crossing)	≈97%

*Company examples identify a specific component or material supplied for that use. Shares refer to the full use category. Shares link to their sources. First use: Ni (2024) and W (2021). Demand or end use: Co and Mo (2025), Cu (2024), Al (2020), Mn (undated). Cr covers ferrochrome (ICDA, 2022). Zr covers zircon output (Iluka, 2024). *Ti covers U.S. metal use (2025), reported as a majority. n.r. = not reported.*

The forms show where each metal enters manufacturing. Ferroalloys go into the melt, while plate, tube, bar and powder go to manufacturers that turn them into components. The profiles below explain how these forms are supplied and used.

4.1 Nickel

Indonesia has become the largest source of nickel over the past decade. The USGS estimates that mines produced 3.90 Mt of contained nickel in 2025, with 2.60 Mt, or about two thirds, from Indonesia. The IEA estimates 4.21 Mt worldwide and puts Indonesia at 63 percent of mining and 45 percent of refining. The totals differ, but both sources show that supply is concentrated (International Energy Agency, 2026b; U.S. Geological Survey, 2026g).

Primary nickel production reached 3.875 Mt, compared with 3.577 Mt of use. Just over half of the primary supply in China and Indonesia was nickel pig iron for stainless steel. Vacuum-melted alloys require class 1 nickel (International Nickel Study Group, 2025, 2026). Class 1 metal is widely available, as shown by the rise in the Chinese share of open LME tonnage from 47 to about 68 percent in the first nine months of 2025. The constraint for vacuum melting is approved feedstock from sources such as Vale in Canada and Glencore Nikkelverk in Norway (International Nickel Study Group, 2025; United Nations Statistics Division, 2026a). US and UK sanctions restrict US imports and exchange warrants for Russian metal produced on or after 13 April 2024 (U.S. Department of the Treasury, 2024).

The nickel surplus helps explain the fall in price. The World Bank recorded an average of USD 25,834 per tonne in 2022 and USD 15,162 in 2025 (World Bank, 2026a). Indonesia first cut its 2026 ore quota to 260 to 270 Mt and later raised it to 290 to 300 Mt. The International Nickel Study Group lowered its 2026 surplus forecast to 19 kt, after a surplus of 299 kt in 2025. Indonesia's quota is therefore an important influence on nickel prices (Argus Media, 2026a; Indonesia Business Post, 2026; International Nickel Study Group, 2026).

Figure 1: The nickel price has fallen by about 41 percent from its 2022 annual average

Source: World Bank Pink Sheet, annual averages of monthly prices, team calculation (World Bank, 2026a, 2026b).

Stainless steel uses most nickel. Gross stainless steel melt shop output reached 64.2 Mt in 2025. The IEA expects nickel demand outside energy technologies, including stainless steel and alloys, to rise from 2,906 kt in 2025 to 3,255 kt in 2035 (International Energy Agency, 2026b; worldstainless, 2026).

Nickel alloys are sold as strip, sheet, plate, bar, wire, and tube. In 2024, Germany exported 17,775 t of nickel plate, sheet, and strip worth USD 578 million, or about USD 32,500 per tonne. Eurostat's volume figure agrees within 2 percent (Eurostat, 2026a; United Nations Statistics Division, 2026c).

Germany's export unit value was 1.9 times the 2024 nickel price. This comparison alone does not measure the value added by conversion because an alloy contains several metals. Alloy C-276 contains about 57 percent nickel, 16 percent molybdenum, and 16 percent chromium. At 2024 and 2025 prices, its metal was worth about USD 19,000 per tonne before counting tungsten. The team estimates that the export unit value was about 1.7 times the value of that metal content (Haynes International, 2023; U.S. Geological Survey, 2026c, 2026f; World Bank, 2026a).

Producers pass metal costs through alloy surcharges or index clauses. ATI does so for most of its 70 million pounds of nickel. Carpenter uses surcharges for sales outside the 43 percent under firm prices and reports sales excluding surcharge (ATI, 2026a; Carpenter Technology,

2026). The base price is therefore important to margins. Special Metals raised its Huntington base prices eight times from April 2021 to February 2024, by 5 to 20 percent each time (Special Metals Corporation, 2024). Product form and qualification help explain why buyers pay above the value of the metal.

4.2 Cobalt

Export restrictions contribute to cobalt scarcity. The USGS estimates that mines produced 310 kt of contained cobalt in 2025. The Democratic Republic of the Congo produced about 230 kt, or 73 percent. China refined 78.6 percent of the world's 240 kt of refined cobalt (Cobalt Institute, 2026; U.S. Geological Survey, 2026d).

In February 2025, Congo banned cobalt exports. It later set an annual quota of up to 96,600 t for 2026 and 2027. CMOC mined 117,549 t in 2025 and received an export allowance of 31,200 t. The European cobalt metal price rose 130 percent during 2025 (CMOC Group, 2026; Cobalt Institute, 2026; Mining.com, 2026; U.S. Geological Survey, 2026d). The IEA expects 2035 mine supply of 225 kt against demand of 300 to 322 kt, which suggests a lasting supply gap (International Energy Agency, 2026b).

Superalloys use a small share of cobalt but require high-purity metal. In 2025, batteries for vehicles and portable electronics used 206 kt, while superalloys used 20 kt. A few Western refineries supply most of the high-purity metal for superalloys (Cobalt Institute, 2026). Cobalt is used in remelt stock for turbine blades and combustors, cobalt-chromium bar and powder for hip and knee implants, and wear-resistant alloys. These uses are difficult to switch to other metals, so a quota set in Kinshasa can raise costs and disrupt production in places such as Toulouse and Cincinnati.

4.3 Chromium

Chromium gives corrosion-resistant and heat-resistant alloys a protective oxide layer. Mining and smelting are concentrated in different countries. World chromite ore output reached an estimated 51 Mt gross weight in 2025. The USGS estimates that South Africa mined about 23 Mt. The International Chromium Development Association reports 25.2 Mt using a different definition (International Chromium Development Association, n.d.; U.S. Geological Survey, 2026c). The association reports that China, which has little chromite, smelts 9.9 Mt of ferrochrome, mostly for stainless steel. The United States imports 79 percent of its chromium needs (International Chromium Development Association, n.d.; U.S. Geological Survey, 2026c; worldstainless, 2026).

Vacuum-melted nickel alloys require more specialized inputs than the ferrochrome used in stainless steel mills. These inputs include low-carbon ferrochrome and aluminothermic chromium metal. The U.S. chromium metal price rose to an estimated USD 5.90 per pound in

2025 while the ferrochrome price fell (U.S. Geological Survey, 2026c). AMG Chrome in the United Kingdom supplies low-gas chromium metal for single-crystal turbine blades. A vacuum melt shop can stop production if it loses a qualified source of this input (AMG Critical Materials N.V., n.d.).

4.4 Molybdenum

Molybdenum helps alloys resist corrosion from hot acids and seawater. The amount needed depends on the grade. One tonne of 316L stainless steel contains about 21 kg of molybdenum. One tonne of Alloy C-276 contains about 160 kg, and one tonne of Alloy 625 contains about 90 kg. These grades can therefore require large amounts of molybdenum even when alloy volumes are small (ATI, n.d.; Haynes International, 2023; Outokumpu, 2025).

Much of the world's molybdenum comes from copper mining. This limits how quickly supply can respond to higher prices. The USGS estimates 2025 mine output at 260 kt of contained molybdenum. The International Molybdenum Association reports 672.6 million pounds, or about 305 kt in the team's conversion. Almost all of the difference is in China (International Molybdenum Association, 2026; U.S. Geological Survey, 2026f).

According to the USGS, only China and the United States produce from primary molybdenum mines. Since February 2025, China has required licenses for molybdenum powder exports. The policy excludes the oxide and ferromolybdenum that melt shops buy (Ministry of Commerce of the People's Republic of China, 2025a; U.S. Geological Survey, 2026f). The U.S. oxide price rose 7 percent to an estimated USD 51 per kg in 2025. European use fell 7 percent as the chemical industry slowed (International Molybdenum Association, 2026; U.S. Geological Survey, 2026f).

4.5 Titanium

Titanium can replace nickel alloys in lighter and cooler aerospace parts. Its mineral and metal markets have very different volumes. In 2025, world ilmenite and rutile output reached 9.8 Mt measured as TiO2, mostly for pigment. Titanium sponge output was 370 kt (U.S. Geological Survey, 2026i, 2026j).

China produced about 260 kt of sponge and held 320 kt of the world's 470 kt of capacity. Western aircraft engines use other qualified sources, and the United States has no operating sponge capacity (U.S. Geological Survey, 2026i). Russia produced about 25 kt of sponge in 2025 and shipped USD 715 million of titanium to the ten reporting countries in 2024. Russia's VSMPO-AVISMA describes itself as the leading supplier of titanium for aviation (U.S. Geological Survey, 2026i; United Nations Statistics Division, 2026a; VSMPO-AVISMA Corporation, n.d.).

The scarce forms are aircraft-grade sponge and the ingot, bar, and plate made from it. Osaka Titanium is increasing aircraft-grade capacity from 40 kt to 50 kt by fiscal 2027. Kobe Steel sells titanium castings and forgings for aircraft, and ATI reports lead times of about 20 months for premium titanium. Europe melts and fabricates titanium without making sponge. In 2024, it sold 22.6 kt of wrought titanium and imported 31.7 kt (ATI, 2026b; Eurostat, 2024; Kobe Steel Ltd., 2026; Osaka Titanium Technologies, 2025). Nickel superalloys face a similar shortage of qualified melting capacity.

4.6 Tungsten

China controls much of the world's tungsten supply. In 2025, mines produced an estimated 85 kt of contained tungsten. China produced about 67 kt, or 79 percent (U.S. Geological Survey, 2026k). China's first 2025 national mining quota covered 58,000 t of concentrate. Since February 2025, it has required export licenses for ammonium paratungstate, tungsten oxides, unsintered carbide, and specified tungsten alloys. Sintered carbide tools and tungsten wire remain outside these controls (Ministry of Commerce of the People's Republic of China, 2025a, 2025b; Ministry of Natural Resources, 2025).

Carbide for cutting and mining tools accounts for 65 percent of first use. Steels and superalloys account for 14 percent (International Tungsten Industry Association, n.d.). In superalloys, tungsten strengthens cast turbine parts and heat-resistant sheet. Export license decisions can therefore affect gas turbine production. Sandvik's powder business more than doubled orders and revenue in the third quarter of 2026 as tungsten prices rose (Sandvik, 2026).

4.7 Other Selected Metals

Zirconium and the minor superalloy elements serve specialized uses. Copper, aluminium, and manganese have much larger markets, with smaller roles in high-performance applications.

4.7.1 Zirconium

World production of zirconium concentrates reached about 1.2 Mt gross weight in 2025. Australia led with 400 kt. Most concentrates go into ceramics, foundry sand, and refractories (U.S. Geological Survey, 2026l). The metal market is much smaller. Fuel makers and integrated producers such as ATI control their nuclear fuel cladding supply. The open market includes zirconium plate and tube for acid service (ATI, 2026a).

4.7.2 Copper

Copper connects high-performance materials to electrification. ICSG and IEA figures put world copper mine production at about 23.2 Mt in 2025. Flat-rolled products account for 11

percent of first use, compared with 63 percent for wire. Copper averaged USD 9,947 per tonne in 2025 and set records above USD 14,500 in 2026 (International Copper Study Group, 2025, 2026; International Energy Agency, 2026b; The Oregon Group, 2026; World Bank, 2026b).

Under current policies, the IEA expects mine supply from current projects to fall to 20.7 Mt by 2035. Demand rises from 27.8 Mt to 32.4 Mt in the same outlook. The IEA estimates a 2035 primary supply deficit of about 25 percent (International Energy Agency, 2026a, 2026b). Specialized forms include copper-nickel and high-conductivity strip. Section 5.6 discusses data center busbars as an example.

4.7.3 Aluminium

Aluminium can replace heavier alloys when temperatures and corrosion conditions allow. World primary production reached about 74 Mt in 2025. China produced 61 percent. European primary production has fallen to about 955 kt, compared with annual demand of about 9 Mt (European Aluminium, n.d.; Federation of Aluminium Consumers in Europe, 2024; International Aluminium Institute, 2026; U.S. Geological Survey, 2026b).

4.7.4 Manganese

Manganese demand mainly follows carbon steel production. Mines produced about 20 Mt of contained manganese in 2025. South Africa, Gabon, Ghana, and Australia supplied about 81 percent of the total. The ore price fell to about USD 4.5 per metric ton unit. The study identifies limited supply risk and growth potential for alloy producers (International Manganese Institute, 2026; U.S. Geological Survey, 2026e).

4.7.5 Additional Metals

Iron is used in alloys such as FeNi36 for display masks and FeNi42 for lead frames. Their value depends mainly on the ability to roll the material to the required specifications.

Niobium and rhenium are used in superalloys. Alloy 718 contains about 5 percent niobium. Brazil mines about 93 percent of the world's niobium, and CBMM sells it as vacuum-grade nickel-niobium. Single-crystal turbine blades contain rhenium. In 2025, mines produced only about 81 t of rhenium worldwide (CBMM, n.d.; RTX, 2026; U.S. Geological Survey, 2026a, 2026h).

5. Industry Needs: Demand, Geography and Market Size

Each of the seven industry sections starts with market size and geography and then explains applications and product forms. The first subsection ends with the best available measure at a named stage of the value chain. Where alloy tonnage is unavailable, chapter 7 uses an index calculated by the team. Appendix A5 lists the underlying indicators.

To avoid counting applications twice, the study groups electrolyzers and fuel cells under energy, process vessels under chemical processing, and sensors and magnetic parts under electronics.

5.1 Automotive

Electrification reduces demand for the heat-resistant nickel alloys used in combustion vehicles. Fully electric cars do not use the same turbochargers, exhaust sensors, glow plugs, and gaskets.

5.1.1 Demand, Geography and Market Size

The world produced 96.4 million motor vehicles in 2025. China produced 34.5 million of them. EU car registrations reached 10.8 million, which was 17 percent below the 2019 level (European Automobile Manufacturers' Association, 2026a; International Organization of Motor Vehicle Manufacturers, 2026).

More than 20 million electric cars were sold in 2025, about a quarter of the market. By 2035, the IEA expects electric cars to make up about half of global sales and more than 90 percent in China and the EU (International Energy Agency, 2026c). The shift does not replace all nickel demand lost from engine alloys. Lithium iron phosphate cathodes contain no nickel and accounted for over 55 percent of electric vehicle batteries in 2025 (International Energy Agency, 2026c).

European policy may slow the shift. In October 2026, two European Parliament committees supported a 95 percent cut in new car CO2 emissions by 2035, or 87.5 percent with credits for low-carbon steel. A plenary vote was due in December 2026 (electrive, 2026).

The best available nickel measure covers battery cathodes. The IEA estimates electric vehicle nickel demand of 439 kt in 2025 and 1,294 kt to 1,536 kt in 2035. Demand for engine alloys is declining, but public data do not show its tonnage (International Energy Agency, 2026b).

5.1.2 Applications and Opportunities

Hybrid and gasoline engines are expected to keep turbochargers in production for another decade. Production reached close to 50 million units in 2025. Garrett expects output to fall toward its 2022 level by 2030 and the share of gasoline engines with turbochargers to peak at 54 percent in 2032 (Garrett Motion, 2026). BorgWarner won three gasoline turbocharger programs in October 2026, with production starting in 2027 (BorgWarner, 2026).

Table 3: High performance metal applications in automotive, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	Current sensors and magnetic shielding in electric drives and chargers	Soft magnetic FeNi strip
Stable to 2030	Turbocharger parts for hybrids and gasoline engines	Heat resistant nickel alloy bar and strip
Falls	Glow plugs, diesel exhaust parts, valves and injectors	Nickel alloys, heat resistant stainless strip

Sources: team assessment (BorgWarner, 2026; Garrett Motion, 2026; International Energy Agency, 2026c).

Implications for suppliers:

- Treat engine alloys for cars as a cash business to 2030 and avoid new capacity for diesel only parts.
- Shift product development toward soft magnetic FeNi strip for sensors and shielding in electric drives.

5.2 Oil and Gas

Oil and gas remains a stable market for high-performance metals. Investment is moving from oil to gas. Sour gas, deepwater production, and liquefied natural gas (LNG) require alloys that can withstand conditions ordinary steel cannot.

5.2.1 Demand, Geography and Market Size

Global upstream oil and gas investment fell from USD 625 billion in 2019 to USD 542 billion in 2025. It was about USD 546 billion in 2026. About 40 percent of investment goes toward slowing the decline of existing fields (International Energy Agency, 2025b, 2026h, 2026j).

North America remained the largest region at USD 165 billion in 2026, down from USD 219 billion in 2019. Middle East investment rose from USD 75 billion in 2019 to USD 93 billion, and its gas investment roughly doubled from USD 22.9 billion to USD 45.4 billion (International Energy Agency, 2026j). Central and South America grew fastest, from USD 42 billion to USD 68 billion. Under current policies, LNG trade could rise from 560 bcm in 2024 to 880 bcm in 2035 (International Energy Agency, 2025c, 2026j). Upstream investment is the best available measure because public sources do not report the tonnes of nickel alloy, duplex, or clad pipe the industry buys.

5.2.2 Applications and Opportunities

Customer approval is the main barrier to entering the oil and gas market. ISO 15156 identifies corrosion-resistant alloys and the conditions in which they can be used in sour service. Producers still need NORSOK M-650 qualification and operator approval for each product form (International Organization for Standardization, 2020; Standards Norway, 2011). Buyers place orders through approved vendor lists maintained by EPC contractors and subsea companies, including SLB and TechnipFMC. TechnipFMC recorded USD 10.1 billion in subsea orders in 2025 (SLB Limited, 2026; TechnipFMC, 2026).

Product forms include nickel alloy bar for wellheads and completion tools, corrosion-resistant well tubing, umbilical tube, and clad pipe. Alleima's hyper duplex SAF 3007 is the successor to super duplex SAF 2507 in umbilical tube. Clad pipe supplier BUTTING reports EUR 880 million in turnover (Alleima, n.d.; BUTTING, 2026).

LNG carriers also use FeNi alloys. Aperam doubled capacity at Imphy, France, in the first half of 2025 to supply shipyards whose membrane tanks use its Invar M93 cryogenic alloy (Aperam, 2025, n.d.). Outokumpu plans to add about 13,000 t of high-nickel flat product capacity at Avesta from 2028. This investment shows the value of supplying approved products even when the overall market is flat (Outokumpu, 2026b).

Table 4: High performance metal applications in oil and gas, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	Clad and lined pipe, subsea equipment, LNG carrier membrane tanks	Nickel alloy strip, plate, welding wire, FeNi36 strip
Stable	Well tools, wellheads, well tubing, valves and fittings for sour gas	Nickel alloy bar, billet, plate, corrosion resistant tubing

Sources: team assessment (Alleima, n.d.; Aperam, 2025; BUTTING, 2026; International Energy Agency, 2026h; TechnipFMC, 2026).

Implications for suppliers:

- Win places on the vendor lists of Middle East and South American operators and of their EPC contractors and subsea majors.
- Grow strip and welding wire for clad pipe makers once NORSOK M-650 qualification is secured.

5.3 Energy and Environmental Technology

Energy technology shows two different trends. Gas turbine makers are struggling to fill orders. Many announced hydrogen projects have yet to generate orders.

5.3.1 Demand, Geography and Market Size

Orders for gas turbines have grown as data centers need power and operators work to keep electricity grids stable. Siemens Energy is sold out until fiscal 2028. It plans to raise annual gas turbine capacity from about 17 GW in fiscal 2024 to more than 30 GW (Siemens Energy, 2025). Investment in gas-fired power rose from USD 64 billion in 2019 to USD 112 billion in 2025. Data centers could use about 945 TWh of electricity in 2030, more than twice their 2024 use (International Energy Agency, 2025a, 2026j).

Low-emissions hydrogen production was about 1 Mt in 2025, compared with total hydrogen demand of more than 100 Mt. The announced 2030 project pipeline fell to 27 Mt (International Energy Agency, 2026d, 2026e).

The IEA database lists 4.3 GW of operating electrolysis capacity and 19.3 GW at final investment decision or under construction. Alkaline electrolyzers use nickel and account for 60 percent of capacity under construction (International Energy Agency, 2026g). Green hydrogen orders at thyssenkrupp nucera, a leading European alkaline electrolyzer maker, fell from EUR 356 million to EUR 26 million in fiscal 2024/25 (thyssenkrupp nucera, 2025).

Nuclear power offers steadier demand that takes longer to develop. Of 81 reactors under construction, 40 are in China. An October 2025 U.S. framework covers at least USD 80 billion of AP1000 reactors (Westinghouse Electric, 2025; World Nuclear Association, 2026b).

The IEA estimates that hydrogen technologies used 2.3 kt of nickel in 2025. Its 2035 estimates reach 5.2 kt under stated policies and 32 kt in the high-demand case. Public sources do not report the tonnage of superalloys used in gas turbines (International Energy Agency, 2026b).

5.3.2 Applications and Opportunities

Table 5 compares when three energy opportunities could generate demand. Gas turbine parts are needed now. Fuel cell demand for specific metals is less clear. Bloom Energy plans to double its annual solid oxide output to 2 GW by the end of 2026, but it does not identify its interconnect material (Bloom Energy, 2026).

In nuclear power, Alleima is reopening its Sandviken tube mill to increase steam generator tube capacity by about 60 percent. It melts and remelts its own alloys there, so the expansion creates little open demand for bar or hollows from other producers (Alleima, 2026a).

Table 5: High performance metal applications in energy technology, outlook to 2035

Outlook	Applications	Materials and product forms
Grows strongly	Gas turbine blades, vanes, combustors and repair	Superalloy remelt stock, combustor sheet, welding wire

Outlook	Applications	Materials and product forms
Grows from 2030	Nuclear steam generator tube	Alloy 690 tube, mostly from integrated mills
Option	Solid oxide cell interconnects, alkaline electrolyser electrodes	Coated ferritic strip, nickel strip and mesh

Sources: team assessment (International Energy Agency, 2026g; Siemens Energy, 2025; thyssenkrupp nucera, 2025; World Nuclear Association, 2026b).

Implications for suppliers:

- Qualify superalloy remelt stock and combustor sheet with turbine casting houses while capacity is sold out to 2028.
- Treat electrolysers, fuel cells and nuclear tube as options released only against signed volumes, and in nuclear tube only where a tube maker buys outside hollows.

5.4 Aerospace

Aerospace is a major source of growth for high-performance metals. Jet engine hot sections use nickel and cobalt superalloys, and suppliers face limits on how quickly they can melt and forge them.

5.4.1 Demand, Geography and Market Size

Airbus expects the passenger fleet to grow from 23,310 aircraft in 2025 to 45,550 in 2045. That forecast calls for 42,060 new aircraft. Boeing expects 43,625 deliveries using a broader definition (Airbus, 2026a; Boeing, 2026). Engine output is growing faster than aircraft output. CFM delivered 1,802 LEAP engines in 2025, up 28 percent (GE Aerospace, 2026a; Safran, 2026a).

Maintenance and repair add another source of demand. Airbus projects that aviation services will grow from USD 209 billion in 2026 to USD 422 billion in 2045. Materials account for more than 85 percent of the cost of maintenance performed off the aircraft wing (Airbus, 2026b).

Defense also adds demand. World military expenditure reached USD 2,887 billion in 2025. GE delivered 635 defense engines, up from 490 a year earlier (GE Aerospace, 2026a; Stockholm International Peace Research Institute, 2026).

Cobalt provides the best measured indicator at the metal stage. The Cobalt Institute estimates that aviation used 11,499 t of the 20 kt of cobalt that went into superalloys in 2025. The U.S. Geological Survey reports that 51 percent of U.S. cobalt consumption goes into superalloys,

mainly for aircraft engines. The tonnage of nickel alloys is not published (Cobalt Institute, 2026; U.S. Geological Survey, 2026d).

5.4.2 Applications and Opportunities

Rotating parts have the highest entry requirements among the applications in Table 6. At plants such as ATI in Lockport and Special Metals in Huntington, disc and shaft billet goes through vacuum induction melting, electroslag remelting, and vacuum arc remelting. The most demanding discs use superalloy powder. Pratt and Whitney's geared turbofan powder metal issue showed how important powder quality is (ATI, 2026a; RTX, 2026; Special Metals Corporation, n.d.).

Engine makers specify approved melt routes in addition to AMS standards and NADCAP processes. GE Aerospace says that qualifying a new supplier takes substantial time. Initial approval for a rotating part can take several years (GE Aerospace, 2026a).

Table 6: High performance metal applications in aerospace, outlook to 2035

Outlook	Applications	Materials and product forms
Grows strongly	Discs, shafts, casings and rings for LEAP, GE9X, Trent and military engines	Triple melt superalloy bar and billet, powder for discs, sheet
Grows	Engine repair and overhaul	Welding wire, repair alloys
Grows	Additively manufactured combustor and nozzle parts	Superalloy powder
Stable	Combustor and wear parts where nickel alloys replace cobalt	Cobalt alloy sheet and plate

Sources: team assessment (Airbus, 2026a; ATI, 2026a; Safran, 2026a).

Major producers respond to shortages by signing supply agreements and building their own capacity. ATI earned USD 1,762.9 million from commercial jet engines in 2025 under long-term agreements with GE Aerospace, Safran, Rolls-Royce, and Pratt and Whitney. ATI reports forging lead times of more than two years (ATI, 2026a, 2026b). Safran committed EUR 150 million to a 30,000 t press at Gennevilliers to produce about 14,000 hot section parts a year from 2029 (Safran, 2026b).

Other suppliers are active as well. Austria's voestalpine Böhler has about EUR 1 billion of aerospace orders over five years. Japan's Proterial sells gas turbine superalloys. Aperam entered the U.S. specialty alloy market by buying Universal Stainless for EUR 415 million in 2025 (Aperam, 2026; Proterial, n.d.; voestalpine AG, 2026). A new qualified European supplier has a chance to enter today, but that chance will narrow as the investments come online around 2030.

Implications for suppliers:

- Start part level qualifications with engine makers and independent forgers now, because rotating approvals take several years.
- Build the repair wire and powder business alongside, because off wing maintenance grows by 3.7 percent a year (Airbus, 2026b).

5.5 Chemical Processing

Chemical processing has long used corrosion-resistant alloys. In Europe, demand now comes mainly from maintenance. Much new plant construction takes place in Asia.

5.5.1 Demand, Geography and Market Size

Europe's chemical industry is producing less and has spare capacity. EU27 chemical sales were about EUR 635 billion in 2024. Europe's share of world sales fell from 21 percent in 2009 to 13 percent, while China's rose from 24 to 46 percent (European Chemical Industry Council, 2025a, 2025b). The EU chemical production index fell 3.2 percent in 2025. Capacity utilization was 75.0 percent in the third quarter of 2026, compared with a long-term average of 81.3 percent (European Chemical Industry Council, 2026b; Eurostat, 2026c).

Investment is following growth to Asia. BASF expects Asia Pacific to account for over 80 percent of chemical industry growth by 2035. It opened its Zhanjiang Verbund site in November 2025 (BASF, 2026). The closest measure at the metal stage is EU apparent consumption of nickel plate, sheet, strip, and foil. This was 57.8 kt in 2024 across all end uses. The share used in chemical processing is unknown (Eurostat, 2025d).

5.5.2 Applications and Opportunities

Corrosion-resistant alloys are needed when stainless steel cannot withstand conditions such as sulfuric acid, hydrochloric acid, or chlor-alkali service. Table 7 shows that European demand comes mainly from replacement. New plants and demand for plate and clad plate are shifting to Asia. Alleima is investing SEK 220 million in further processing and finishing at Mehsana, India. Its chemical and petrochemical product range includes Sanicro 625 (UNS N06625) tube with an outside diameter of 6 to 42 mm (Alleima, 2025, 2026b).

Table 7: High performance metal applications in chemical processing, outlook to 2035

Outlook	Applications	Materials and product forms
Stable	Replacement heat exchangers, vessels and piping in European plants	C-276, C-22 and 59 type NiCrMo alloys and Alloy 625 as plate, sheet and welding consumables

Outlook	Applications	Materials and product forms
Grows	New plants in China, India and Southeast Asia	Plate, clad plate, zirconium plate and tube
Selective	Battery material plants, where titanium and linings dominate	Plate for selected duties, welding consumables
Falls	New build in European base chemicals	Plate

Sources: team assessment (Alleima, 2026b; BASF, 2026; European Chemical Industry Council, 2026b).

Implications for suppliers:

- Defend European replacement demand with plate and welding consumables sold on documented life cycle cost.
- Follow new construction to India and Southeast Asia through the engineering contractors that build Asian plants.

5.6 Electronics and Electrical Engineering

Electronics has the largest headline market figures but the smallest metal volumes. Chip and data center applications use specialized strip, foil, and high-purity alloys in amounts measured in tonnes.

5.6.1 Demand, Geography and Market Size

Semiconductor sales reached USD 795.6 billion in 2025, up 26.2 percent. WSTS forecasts USD 1.51 trillion in 2026, mainly because it expects memory sales to more than triple. Higher sales value does not imply a similar rise in metal volume (World Semiconductor Trade Statistics, 2026).

Equipment spending is a closer indicator of alloy demand. SEMI expects spending on 300 mm fab equipment to reach USD 149 billion in 2026 and USD 227 billion in 2029. Fabs are located in Taiwan, Korea, China, and increasingly in the United States (SEMI, 2026a, 2026c). Public sources do not report electronics demand in tonnes of FeNi or high-purity alloy. FeNi growth comes from electronics and electric drive sensors, while demand from engine alloys follows a different trend.

5.6.2 Applications and Opportunities

NVIDIA says that a 1 MW data center rack operating at 54 VDC needs up to 200 kg of copper busbar. It plans to use 800 VDC distribution with Kyber racks in 2027. Its partners include

Infineon, Vertiv, Eaton, and Schneider Electric. Vertiv raised its copper hedge to 12,946 t by September 2026 (NVIDIA, 2025a; Vertiv, 2026).

Dai Nippon Printing (DNP) doubled its OLED metal mask capacity with an 8th-generation line at Kurosaki. It expects its invar patents to remain protected (Dai Nippon Printing, 2024a, 2024b). DNP notes that some panel makers are testing mask-free deposition. TrendForce reports that Visionox and TCL CSOT are pursuing this method, while Samsung Display and BOE continue to use fine metal masks on 8.6-generation lines (Dai Nippon Printing, 2024b; TrendForce, 2025).

Table 8: High performance metal applications in electronics, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	OLED fine metal masks for IT panels, at risk from mask free patterning	FeNi36 ultra thin foil
Grows	Current sensors and shielding in power electronics and data centres	Soft magnetic FeNi strip
Grows	Fab gas delivery and chamber components	High purity nickel alloy and stainless strip and tube stock
Stable	Glass seals, lead frames, resistance heating	FeNi42 and FeNi29Co18 strip, resistance wire

Sources: team assessment (Dai Nippon Printing, 2024a; Infineon Technologies, 2026; NVIDIA, 2025a; SEMI, 2026a).

The filings of TSMC, Applied Materials, and Lam Research do not name specific alloys. Alloy suppliers therefore reach these customers through component makers. Infineon's EUR 5 billion Smart Power Fab, which opened in Dresden in July 2026, shows that Europe is still adding power electronics capacity (Infineon Technologies, 2026).

Implications for suppliers:

- Partner with mask makers to qualify FeNi36 ultra thin foil rather than compete with protected processes.
- Qualify high purity alloy strip and tube with the component suppliers of fab gas systems.

5.7 Medical Technology

Medical technology offers the steadiest growth of the seven industries. Aging populations drive demand for hip and knee replacements made from cobalt-chromium, titanium, and zirconium alloys. This demand is less closely tied to business cycles.

5.7.1 Demand, Geography and Market Size

Joint replacements are the most measurable driver of implant metal demand. In 2023, the OECD average was 198 hip and 156 knee replacements per 100,000 people, after a decade of rising rates (OECD, 2025). Europe had a medical technology market of about EUR 173 billion in 2025. The United States accounted for 46.4 percent of global device sales in 2024 (MedTech Europe, 2025, 2026). Central purchasing in China cut artificial hip and knee prices by about 82 percent, limiting profits from implant metals despite the market's size (National Healthcare Security Administration, 2022).

The best measure is the number of procedures. In 2023, the 35 OECD members reporting both procedures recorded about 2.75 million hip and knee replacements, excluding the United States and Japan. Implant alloy tonnage is not published (OECD, 2025).

5.7.2 Applications and Opportunities

A few large companies buy most implant metals, so approval and reliability are especially important. Stryker reported USD 9,469 million in orthopedic revenue in 2025. The four leading orthopedic groups had more than USD 20 billion in implant revenue (Johnson & Johnson, 2025; Smith & Nephew, 2026; Stryker, 2026). Demand for mills can still change quickly. Carpenter's medical sales excluding surcharge fell 24 percent to USD 224.3 million in fiscal 2026 as customers reduced inventories (Carpenter Technology, 2026).

ASTM standards differ by product form. ASTM F75 covers cast CoCrMo, including investment castings such as knee femoral components and casting alloy sold as ingot or bar. ASTM F1537 covers wrought CoCrMo bar, rod, and wire. ASTM F3213 covers CoCrMo made by powder bed fusion. One example is the printed talus implant that the FDA approved in 2023 (ASTM International, 2017, 2020, 2023; U.S. Food and Drug Administration, 2023).

Smith & Nephew's OXINIUM uses an oxidized surface on a Zr2.5Nb alloy as an alternative to cobalt-chromium femoral components. It reaches 94.1 percent survivorship at 20 years. It can replace some CoCr demand, but it requires wrought Zr2.5Nb bar that meets ASTM F2384. The supplier is undisclosed (ASTM International, 2024, n.d.; Smith & Nephew, 2026).

Table 9: High performance metal applications in medical technology, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	Cast knee femoral components	CoCrMo casting alloy, ingot and bar to ASTM F75
Grows	Wrought hip and knee components	CoCrMo bar to ASTM F1537
Grows	Additively manufactured implants	CoCrMo powder to ASTM F3213

Outlook	Applications	Materials and product forms
Grows	Oxidized zirconium femoral components	Zr2.5Nb bar to ASTM F2384
Falls	Implants for China under central purchasing	CoCr bar

Sources: team assessment (ASTM International, 2023; National Healthcare Security Administration, 2022; Smith & Nephew, 2026).

Implications for suppliers:

- Offer CoCrMo casting alloy to ASTM F75 and bar to ASTM F1537 to implant makers in Europe and the United States.
- Develop CoCrMo powder to ASTM F3213 and Zr2.5Nb bar to ASTM F2384, and pass the supplier audits of the device makers.

6. Global Market Size and Trade

Trade data help estimate the market for qualified metal products and show where conversion takes place. This chapter examines market size, major trade routes, and policies that affect those routes.

6.1 Measuring the Market for Qualified Nickel Products

Official statistics allow the study to estimate apparent consumption for nickel mill products in Europe. Apparent consumption is production plus imports minus exports. In 2024, the EU consumed an estimated 57.8 kt of nickel plate, sheet, strip, and foil and 9.2 kt of tubes. The bar estimate depends on a confidential production figure (Eurostat, 2025a, 2025b, 2025d).

Eurostat's reported production values fall below the value of the nickel they contain. Table 10 therefore applies EU border unit values to the volumes. This gives an estimated EUR 1.8 to 2.8 bn for flat products and EUR 0.24 to 0.38 bn for tubes (Eurostat, 2025c). These codes only cover alloys in which nickel is the main metal, so Table 10 captures only part of the qualified market.

Table 10: European market for nickel mill products, 2024

Product code	EU apparent consumption	Extra EU unit value, imports and exports (EUR/kg)	Value, team estimate
Bars, rods and wire (PRODCOM 24452100)	Confidential	25.4 and 33.4	Not estimated

Product code	EU apparent consumption	Extra EU unit value, imports and exports (EUR/kg)	Value, team estimate
Plate, sheet, strip and foil (24452200)	57.8 kt	47.8 and 31.0	EUR 1.8 to 2.8 bn
Tubes and fittings (24452300)	9.2 kt	41.1 and 26.1	EUR 0.24 to 0.38 bn
Contained nickel, 2024		15.5	

Sources: Eurostat PRODCOM DS-059358, apparent consumption as sold production plus extra EU imports minus exports, team calculation of unit values and value range (Eurostat, 2025c, 2026d). Contained nickel from the World Bank price at the ECB rate (European Central Bank, 2025; World Bank, 2026a).

6.2 Conversion Creates the Trade Surplus

Germany leads the ten reporting countries in exports of nickel plate, sheet, and strip. It ranks second for bars and wire. Its exports across the three mill product codes rose from USD 0.968 bn in 2019 to USD 1.418 bn in 2024. This pattern shows the value of converting nickel into mill products (United Nations Statistics Division, 2026b, 2026c, 2026d).

Table 11: Leading exporters of nickel mill products and unwrought nickel, 2024

HS code and product	Leading exporters among the ten reporters (USD million)	Germany net trade (USD million)
7505 Nickel bars, rods, profiles, wire	United States 1,400, Germany 638, France 428, United Kingdom 402, Italy 388 (mirror: Austria 384)	+79
7506 Nickel plate, sheet, strip, foil	Germany 578, United States 482, Japan 274, France 110, China 85 (mirror: Sweden 101)	+414
7507 Nickel tubes and fittings	United States 324, Germany 202, China 195, United Kingdom 168, Italy 141	+105
7502 Unwrought nickel	China 2,068, Indonesia 862, United Kingdom 768 (mirror: Canada 1,236, Norway 1,086, Russia 1,049, Australia 981)	−736

Sources: UN Comtrade reporter data for Germany, the United States, China, Japan, Korea, India, France, the United Kingdom, Italy and Indonesia (United Nations Statistics Division, 2026e). Mirror values are these ten countries' imports from a non reporter, a lower bound (United Nations Statistics Division, 2026a). Net trade is a team calculation.

Germany imports USD 736 million more raw nickel than it exports. It is a net exporter of every nickel mill product in Table 11. Its plate and strip show the conversion premium described in section 4.1. EU export values show a similar pattern, at EUR 31.0 per kg compared with EUR 15.5 for the contained nickel (Eurostat, 2025c; United Nations Statistics Division, 2026c; World Bank, 2026a).

Data from the reporting countries alone suggest that China leads unwrought nickel exports. Import data from their trading partners give a fuller picture. These mirror data show that Canada, Norway, Russia, and Australia each shipped about USD 1 bn to the ten reporting countries. These are major sources of class 1 nickel, and each shipped more than Indonesia reports (United Nations Statistics Division, 2026a). The same data identify Austria and Sweden as converters, similar to Germany. Section 6.3 follows the trade routes that connect these producers to their buyers.

Figure 2: German exports of nickel mill products rose by about 47 percent from 2019 to 2024

Source: UN Comtrade, Germany as reporter, exports to the world in current USD (United Nations Statistics Division, 2026b, 2026c, 2026d).

6.3 Corridors and Dependencies

The main trade routes reflect where buyers operate. Germany exported about USD 111 to 112 million of nickel bar and wire to each of the United Kingdom and the United States in 2024. These are key aircraft engine markets described in section 5.4. Germany's plate and strip exports go mainly to Austria, Italy, China, and the United States, where chemical and electronics fabricators operate (United Nations Statistics Division, 2026b, 2026c). British exports of turbojets and gas turbines reached USD 39.8 bn in 2024, including re-exports (United Nations Statistics Division, 2026e).

6.4 Trade Policy Reshapes Sourcing and Access

Trade policy can change costs and market access quickly. In 2025, the United States raised Section 232 tariffs on steel and aluminium to 50 percent. From 1 August 2025, it taxed the copper content of semifinished copper products at 50 percent. Its tariff schedule lists general duties of 2 to 3 percent for nickel bars, plate, strip, and tubes under HS 7505 to 7507 (The White House, 2025; United States International Trade Commission, 2026).

The EU Carbon Border Adjustment Mechanism entered its definitive phase on 1 January 2026. It covers ferrochrome and ferronickel and excludes nickel products in HS chapter 75. It can therefore raise costs for inputs without protecting nickel alloy mill products (Ernst & Young, 2025). The EU Critical Raw Materials Act lists nickel, cobalt, and tungsten as strategic or critical (European Parliament and Council, 2024). Together with the quotas and licenses in

chapter 4, these measures give producers a reason to convert metals close to customers and qualify several sources for each critical input.

7. Growth Scenarios to 2035

This chapter estimates alloy demand through 2035. The team used a calibrated tier 1 model for established applications and a separate tier 2 register for emerging applications (Product Strategy 2035 Team, 2026a).

7.1 Growth Model and Scenario Assumptions

Tier 1 models demand as D = A × I × S. A represents industry activity, I represents alloy mass per unit, and S represents the share of units using a high-performance alloy. Each factor is a ratio of its 2035 value to its 2025 value. The index sets 2025 at 100. The model uses measured end uses, including the 11,499 t of cobalt used in aviation out of 20.0 kt used in superalloys in 2025. The IEA outlook for nickel outside energy technologies sets the broader limit for the estimates (Cobalt Institute, 2026; International Energy Agency, 2026b).

Table 12: Tier 1 drivers and assumptions, low to high range

Industry	Activity driver	Growth range to 2035	Intensity, share	Key source
Aerospace	New engines and aftermarket	2.0 to 5.0% and 3.0 to 4.2% a year	Share 0.90 to 1.03	(Airbus, 2026b)
Energy	Gas turbines, nuclear, other	Turbine output ratio 1.15 to 1.85	±1% a year	(Siemens Energy, 2025)
Electronics	300 mm fab capacity, components	Index 138 in 2031, then 0 to 6%	−1.5 to +1% a year	(SEMI, 2026b)
Medical	Hip and knee procedures	3.0 to 4.1% a year	CoCr share 0.90 to 1.02	(OECD, 2025)
Chemical	Chemical activity	−1.0 to 2.5% a year	±0.5% a year	(European Chemical Industry Council, 2026b)
Oil and gas	Upstream investment, LNG	Ratio 0.80 to 1.12 and 1.15 to 1.51	−1.5 to +1% a year	(International Energy Agency, 2026h)
Automotive	Car sales, electric share	Electric share 40 to 62% in 2035	Turbo alloy 0 to 2% a year	(International Energy Agency, 2026c)

Source: team assumptions, all parameters in Table A1.

The team used a backtest to see how well past trends predicted recent outcomes. A trend fitted through 2019 underestimated gas power investment from 2020 to 2025. It had a mean absolute percentage error of 36.5 percent and a bias of minus 36.0 percent. The trend for EU vehicle output overestimated output by 23.0 percent (Product Strategy 2035 Team, 2026a).

The model corrects half of each growth error, up to 1 percentage point per year. The remaining error widens the range in the Monte Carlo simulation. The team used 10,000 draws and seed 20261108 to estimate P10, P50, and P90, the 10th, 50th, and 90th percentiles.

7.2 Results by Industry, Product, and Region

Figure 3 and Table 13 show growth in turbines, engines, electronics, and implants. The automotive index stays near 98. Its two components move in different directions: combustion parts fall to 66 and electric vehicle parts rise to 192 (Product Strategy 2035 Team, 2026a).

Figure 3: Energy technology and aerospace lead alloy demand to 2035, while automotive stays flat

Source: team calculation, band P10 to P90, line P50 (Product Strategy 2035 Team, 2026a).

Table 13: Tier 1 alloy demand index 2035 by industry (2025 = 100)

Industry	P10	P50	P90	P50 growth a year
Energy	126	156	193	4.5%
Aerospace	137	147	158	3.9%
Electronics	112	141	176	3.5%
Medical	132	140	149	3.4%
Oil and gas	93	115	143	1.4%
Chemical	87	109	135	0.9%
Automotive	81	98	119	−0.2%

Source: team calculation (Product Strategy 2035 Team, 2026a).

The model projects stronger demand growth for precision FeNi and vacuum-melted bar than for flat products and tubes. The aerospace weight of 0.575 comes from measured data. The team estimated the other weights (Product Strategy 2035 Team, 2026a).

Table 14: Product family index 2035 (2025 = 100) and main industry weights

Product family	Main industry weights	P10	P50	P90
Precision FeNi strip and foil	Electronics 0.60, electric vehicles 0.30	134	157	185
Superalloy bar, billet, remelt stock	Aerospace 0.575, energy 0.30	131	143	158
Nickel alloy bar, rod and wire	Chemical 0.25, oil and gas 0.25	115	128	142
Plate, sheet, strip and foil	Chemical 0.30, energy 0.20	113	125	139
Tubes and fittings	Oil and gas 0.35, chemical 0.25	111	125	141

Source: team calculation, weighted industry indices (Product Strategy 2035 Team, 2026a).

The supply analysis estimates that IEA cobalt mine supply falls 30 percent short of 2035 demand. The gap reaches 36 percent if the Congo quota stays at its 2026 level. Superalloys require only 13 to 20 percent of the cobalt, so the shortfall is more likely to raise costs than to cap superalloy production. The estimated 21 percent refined nickel gap is an upper bound. Copper mines cover 62 percent of projected demand before scrap is counted (Product Strategy 2035 Team, 2026a; U.S. Geological Survey, 2026d).

Engine and turbine growth is concentrated in the United States and Western Europe. The United States accounts for a fifth of superalloy cobalt. China has 40 of the 81 reactors under construction, so much reactor tube demand is tied to China. Asia accounts for more than 80 percent of expected chemical industry growth (BASF, 2026; U.S. Geological Survey, 2026d; World Nuclear Association, 2026b).

Table 15 lists emerging opportunities, some drawn from the team's research workbook. The team adds an opportunity to tier 1 only after its stated condition is met (Product Strategy 2035 Team, 2026a, 2026b).

Table 15: Tier 2 register, not included in tier 1

Opportunity, alloy and form	TRL	Evidence	Earliest volume	Signpost
Solid oxide cells, ferritic strip	9	Early signal	2026	Strip contract at Bloom Energy
Electrolysers, nickel strip and mesh	8 to 9	Early signal	2026	Nucera orders above EUR 356 m
Additive engine parts, superalloy powder	7 to 9	Early signal	2027	Part specification with volume
800 VDC racks, less copper busbar	6 to 7	Early signal	2027	Eaton and Vertiv busbar launches
Small reactors, zirconium, Alloy 690 tube	7 to 8	Early signal	2030	Serial tube order after licence
Hydrogen turbines, castings and sheet	5 to 7	Speculative	2030	Order with hydrogen supply
Fusion magnets, Alloy C-276 tape foil	4 to 6	Speculative	2030	Magnet order for a pilot plant
sCO2 recuperators, nickel alloy plate	5 to 6	Speculative	2032	Pilot above 10 MW

Sources: team register, TRL and years are team estimates (Bloom Energy, 2026; NVIDIA, 2025b; Product Strategy 2035 Team, 2026a, 2026b).

Chapter 8 scores the measured opportunities in four growing industries and the emerging applications with dated milestones.

8. Strategic Recommendations

The previous chapters examined the market from raw materials through demand growth. This chapter uses those results to rank opportunities for producers of high-performance metals.

8.1 Priority Products, Industries, Regions, and Customers

Table 16 scores both tiers from the perspective of a qualified producer entering or expanding in a segment. Scores range from 1 to 5, with 5 the most attractive. Market size counts for 15 percent, growth for 20 percent, and supply security for 15 percent. Each remaining criterion

counts for 10 percent. Table A4 explains the scoring rules. For example, the growth score uses the P50 growth bands in Table 13. Opportunities supported only by early signals must meet further conditions before producers commit to them.

Table 16: Opportunity matrix, weighted scores (1 = low, 5 = high attractiveness)

Opportunity	Ev	Mkt	Gro	Sup	Comp	Tech	Cust	Barr	Diff	Score	Priority
Turbine remelt stock and sheet	M	4	5	3	2	4	5	5	4	4.05	1
Engine bar and billet	M	5	4	3	2	5	5	4	4	4.00	1
FeNi36 foil and FeNi strip	M	3	5	5	2	5	3	3	4	3.90	1
CoCr implant bar and powder	M	3	4	2	3	4	4	5	3	3.45	2
Sour service bar, strip, wire	M	4	2	4	3	3	3	5	3	3.30	2
Chemical plate, welding wire	M	5	2	4	2	3	3	3	3	3.15	2
Alloy 690 bar and hollows	M	2	2	5	3	4	4	3	3	3.15	2
Solid oxide cell strip	E	2	3	4	3	4	2	3	4	3.10	Option
Alloy C-276 fusion tape foil	E	1	2	4	4	5	2	3	5	3.05	Option
Additive engine powder	E	2	3	3	2	4	4	3	3	2.95	Monitor
Car exhaust and turbo alloys	M	3	1	5	3	1	2	3	2	2.50	Harvest

Source: team assessment, rules in Table A4 (Product Strategy 2035 Team, 2026a). Ev evidence (M measured, E early signal), Mkt market, Gro growth, Sup supply security, Comp competitive intensity (5 = low), Tech technology, Cust customers, Barr entry barriers, Diff differentiation. Priority 1 from 3.75, 2 from 3.00.

The highest-ranked opportunities include vacuum-melted superalloys in the United States and Europe. Gas turbines need remelt stock and combustor sheet for Siemens Energy and casting houses such as CPP. Aircraft engine forgers that supply GE Aerospace and Safran need triple-melt Alloy 718 and 720Li billet.

The main constraint is qualified capacity. Siemens Energy is sold out to 2028, and ATI reports forging lead times beyond two years. Established suppliers such as ATI, PCC, Carpenter, Aubert & Duval, and Daido hold long-term agreements. The competition score is therefore 2, and entrants can win business as qualified second sources (ATI, 2026b; Hepher et al., 2026; Siemens Energy, 2025).

East Asian FeNi36 mask foil for DNP and BOE also ranks in priority 1, with a P50 of 157. Maskless patterning could replace invar foil in some applications (Display Daily, n.d.).

Priority 2 covers products that require customer approval. They include CoCr bar for Stryker and Zimmer Biomet, sour-service products in the Americas, chemical plate sold through Asian fabricators, and Alloy 690 for tube makers (Stryker, 2026; Zimmer Biomet, 2026). All three priority 1 opportunities keep their ranks when the criteria receive equal weights or when entry barriers receive a 20 percent weight.

8.2 Recommended Actions and Implementation Roadmap

Producers should add capacity after customers commit to orders. Table 17 identifies who leads each action, compares peer investments, and states the conditions for moving forward.

Table 17: Implementation roadmap 2026 to 2035

Initiative, owner	2026 to 2028	2029 to 2035	Benchmark and gate
1. Superalloy, sales and quality	Qualify remelt stock and 718 billet	VIM and VAR capacity	ATI Monroe VIM 9 kt a year, Carpenter Athens capex USD 243 m. Gate: contracts for half by 2028
2. FeNi, development	Qualify foil with mask makers	Thin rolling line	Aperam Imphy VIM, EUR 160 m plan. Gate: approval by 2028
3. Franchises, sales	Keep NORSOK, F1537, ASME III	Stock in India	Alleima Mehsana SEK 220 m. Gate: Asian orders rise
4. Tier 2 options, technology	Pilot lots	Scale after signpost	Carpenter powder margin 7.9%. Gate: Table 15
5. Inputs, procurement	Several cobalt sources	Closed loop revert	Daido JPY 36 bn. Gate: no source above half

Sources: company report benchmarks (Alleima, 2026b; Aperam, 2026; Argus Media, 2026b; Carpenter Technology, 2026; SteelOrbis, 2026).

The study recommends dropping investments in parts used only in diesel engines and new European base chemical plants. It also recommends holding electrolyzer capacity until orders are funded and avoiding captive zirconium for nuclear cladding. Appendix A8 gives an early warning and a stopping condition for each risk.

9. Conclusion

The study asked how supply concentration, metal conversion, and industry needs shape opportunities through 2035. The findings show that converting metals into qualified products creates much of the value. Nickel is in surplus, while quotas and licenses restrict cobalt, tungsten, and molybdenum. Countries that make qualified bar, strip, and tube earn trade surpluses on those products. Cobalt shortages are more likely to raise superalloy costs than to cap production.

The tier 1 results reported here put the 2035 P50 index at 156 for energy technology and 147 for aerospace. Automotive demand stays close to its starting level. The highest-ranked opportunities supported by measured evidence are turbine remelt stock, engine billet, and precision FeNi foil in the United States, Europe, and East Asia. Products that require customer

approval follow. The tier 2 register tracks solid oxide strip, fusion magnet tape, additive powder, and small reactor tubing against milestones.

The estimates have limits. Official statistics stop at the mill product stage, and most indices rely on activity measures and weights estimated by the team because alloy tonnage by industry is unavailable. The results support three actions through 2035. Producers should secure their inputs, qualify early with engine and turbine makers, and invest in capacity after customers sign orders.

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Appendices: Data, Sources, Calculations, and Assumptions

A1 Definitions and the stage ladder

Mt means million tonnes, kt means thousand tonnes, and t means metric tonnes. GW means gigawatts, and bcm means billion cubic meters. Prices are in current U.S. dollars or euros for the year stated.

The value chain has five separate stages: contained metal in mine output, refined metal or intermediates, alloy melt, mill products, and finished components. Examples of intermediates are nickel pig iron and cobalt hydroxide. Mill products include strip, sheet, plate, bar, and wire. The study reports stages separately. It also separates contained metal from gross weight, actual output from capacity, and commodity benchmark prices such as the World Bank Pink Sheet from prices for qualified alloy products.

A2 Outlook model, backtest and opportunity scoring

This appendix documents the formulas, parameters, backtest, and simulation settings for the tier 1 model in chapter 7. It also explains the scoring rules used in chapter 8. Values are team estimates unless a source is listed.

To reproduce the results, run python3 model/outlook_2035.py from the repository root. The script updates data/clean/model_tier1_results.csv, model_backtest.csv, model_calibration.csv, model_supply_overlay.csv, model_tier2_register.csv, research/model/outlook_results.md, and Figures 3 and 4 with the same numbers. Use --no-log to skip updating the source register (Product Strategy 2035 Team, 2026a).

A2.1 Formulas and simulation settings

The model divides each industry j into components k. For example, the aerospace components include new engines and the aftermarket. The 2035 index is I(j) = 100 × Σ w(k) × A(k) × (1 + i(k))^10 × S(k) × exp(10 × (Δg(j) + σ(j) × ε)). In this formula, A is the activity ratio from 2035 to 2025, i is the annual trend in alloy mass per unit, S is the ratio of the share using high-performance alloy, and w is the component weight. Δg is the backtest adjustment, σ carries the backtest error into the range, and ε is a standard normal draw.

The adjustment is Δg = −λ × e, where e is the mean annual growth error for the industry's real backtest series. The team set λ = 0.5 and capped the adjustment at 1 percentage point per year. The error term σ equals κ times the root mean square growth error across the industry's series. The team set κ = 0.5 and capped the error term at 3 percentage points.

A product family index is the weighted sum of the industry indices. Table A1 lists the weights. To estimate cobalt tonnes, the model multiplies the 2025 base by the index and by the cobalt content ratio.

The simulation uses 10,000 draws and random seed 20261108. It draws values from triangular distributions with the low, mode, and high values shown in Table A1. A Gaussian copula links activity drivers using a global cycle weight of 0.2 and an industry weight of 0.3. The model draws intensity and share independently. Component and product weights vary around their most likely values in a Dirichlet distribution with concentration 40.

These settings, λ, κ, and both caps are team assumptions. The team used assumptions because the backtest covers short periods and includes the pandemic.

A2.2 Parameters and their sources

Table A1 lists each tier 1 parameter, its range, and its evidence. When a parameter has no cited source, it is a team assumption. The reasons appear in research/model/outlook_results.md.

Table A1: Tier 1 model parameters, low / mode / high

Industry	Parameter	Low / mode / high	Basis and source
Aerospace	Aftermarket growth a year	3.0 / 3.7 / 4.2%	Sourced (Airbus, 2026b; Boeing, 2026; International Civil Aviation Organization, n.d.)
Aerospace	Engine output ratio 2026 to 2025	1.16 / 1.18 / 1.19	Sourced, LEAP high teens growth (GE Aerospace, 2026b; Safran, 2026a)
Aerospace	Engine output growth 2026 to 2035 a year	2.0 / 3.4 / 5.0%	Airbus fleet path, range team assumption (Airbus, 2026a)
Aerospace	Superalloy mass per engine, trend a year	−1.0 / 0.0 / 1.0%	Team assumption (Airbus, 2026a)
Aerospace	Superalloy share of hot section, ratio	0.90 / 0.98 / 1.03	Team assumption, ceramic composites and TiAl
Aerospace	Weights new engines and aftermarket	0.5 and 0.5	Team assumption (GE Aerospace, 2026a)
Energy	Gas turbine output ratio	1.15 / 1.58 / 1.85	Siemens Energy path 17 to 30 GW (International Energy Agency, 2026j; Siemens Energy, 2025, 2026)
Energy	Nuclear fleet and tube ratio	1.05 / 1.20 / 1.45	Derived (International Atomic Energy Agency, 2026; Westinghouse Electric, 2026; World Nuclear Association, 2026a)
Energy	Other equipment growth a year	2.0 / 3.0 / 3.6%	Electricity demand proxy, team assumption

Industry	Parameter	Low / mode / high	Basis and source
Energy	Alloy mass per GW, trend a year	−1.0 / 0.0 / 1.0%	Team assumption
Energy	Alloy share of hardware, ratio	0.97 / 1.00 / 1.03	Team assumption
Energy	Weights turbines, nuclear, other	0.55, 0.25, 0.20	Team assumption
Electronics	Realisation of SEMI path to 2031	0.85 / 1.00 / 1.05	SEMI index 138 in 2031 (SEMI, 2026b)
Electronics	Capacity growth 2031 to 2035 a year	0.0 / 3.0 / 6.0%	Team assumption (SEMI, 2026b)
Electronics	Components and masks growth a year	0.0 / 2.5 / 6.0%	Team assumption (Dai Nippon Printing, 2024b; ZVEI, 2026)
Electronics	Alloy per unit, trend a year	−1.5 / 0.0 / 1.0%	Team assumption (NVIDIA, 2025a)
Electronics	Alloy share, ratio	0.95 / 1.00 / 1.05	Team assumption (Dai Nippon Printing, 2024b)
Electronics	Weights equipment and components	0.6 and 0.4	Team assumption (SEMI, 2026a)
Medical	Hip and knee procedures growth a year	3.0 / 3.5 / 4.1%	Derived, 30 OECD countries 2013 to 2024 (OECD, 2025)
Medical	Other medical uses growth a year	1.8 / 3.0 / 5.6%	MedTech Europe value growth (MedTech Europe, 2025, 2026)
Medical	Alloy per procedure, trend a year	−0.5 / 0.0 / 0.5%	Team assumption
Medical	CoCr and nickel share, ratio	0.90 / 0.97 / 1.02	Team assumption, oxidised zirconium and titanium
Medical	Weights implants and other	0.8 and 0.2	Team assumption (Stryker, 2026)

Industry	Parameter	Low / mode / high	Basis and source
Chemical	Activity growth a year	−1.0 / 1.0 / 2.5%	Team assumption (BASF, 2026; European Chemical Industry Council, 2026a, 2026b)
Chemical	Alloy per output, trend a year	−0.5 / 0.0 / 0.5%	Team assumption
Chemical	Alloy share, ratio	0.97 / 1.00 / 1.03	Team assumption
Oil and gas	Real upstream investment ratio	0.80 / 1.00 / 1.12	IEA scenarios (International Energy Agency, 2025c, 2026h)
Oil and gas	LNG trade ratio	1.15 / 1.35 / 1.51	IEA, 560 to 880 bcm (International Energy Agency, 2025c, 2026f)
Oil and gas	Alloy per unit, trend a year	−1.5 / −0.5 / 1.0%	Team assumption, clad pipe saves material
Oil and gas	Corrosion resistant alloy share, ratio	0.97 / 1.00 / 1.03	Team assumption
Oil and gas	Weights upstream and LNG	0.7 and 0.3	Team assumption (International Energy Agency, 2026i)
Automotive	New car sales ratio	0.95 / 1.00 / 1.10	Team assumption (European Automobile Manufacturers' Association, 2026a; International Energy Agency, 2025d)
Automotive	Electric share of sales 2035	40 / 50 / 62%	IEA scenarios (International Energy Agency, 2025d, 2026c)
Automotive	Heat resistant alloy per combustion car, trend a year	0.0 / 1.0 / 2.0%	Team assumption (Garrett Motion, 2026)
Automotive	Nickel alloy share of turbo parts, ratio	0.95 / 1.00 / 1.03	Team assumption

Industry	Parameter	Low / mode / high	Basis and source
Automotive	Precision alloy per electric car, trend a year	−1.0 / 0.0 / 2.0%	Team assumption
Automotive	FeNi share of motor and sensor parts, ratio	0.90 / 1.00 / 1.10	Team assumption
Automotive	Weights thermal and electric vehicle parts	0.75 and 0.25	Team assumption (Garrett Motion, 2026)
All	Nickel content of alloy class, ratio	0.98 / 1.00 / 1.02	Team assumption
All	Cobalt content, aerospace and other, ratio	0.92 / 1.00 / 1.04 and 0.90 / 1.00 / 1.03	Team assumption (CMOC Group, 2026)
Families	Superalloy bar, billet and remelt stock	Aerospace 0.575, energy 0.300, car thermal 0.050, oil and gas 0.040, chemical 0.035	Aerospace weight measured (Cobalt Institute, 2026)
Families	Nickel alloy bar, rod and wire	Chemical 0.25, oil and gas 0.25, energy 0.20, aerospace 0.15, medical, car thermal, electronics 0.05 each	Team assumption (Alleima, 2026a)
Families	Plate, sheet, strip and foil	Chemical 0.30, energy 0.20, aerospace 0.20, oil and gas 0.15, car thermal 0.10, electronics 0.05	Team assumption (ATI, 2026a)
Families	Tubes and fittings	Oil and gas 0.35, chemical 0.25, energy 0.25, aerospace, car thermal, medical 0.05 each	Team assumption (Alleima, 2026a; International Atomic Energy Agency, 2026)
Families	Precision FeNi strip and foil	Electronics 0.60, electric vehicle parts 0.30, medical 0.10	Team assumption (Dai Nippon Printing, 2024b; International Energy Agency, 2026c)

Sources: model/outlook_2035.py and research/model/outlook_results.md (Product Strategy 2035 Team, 2026a). The calibration in data/clean/model_calibration.csv reproduces the 20.0

kt of superalloy cobalt in 2025 exactly and leaves the IEA nickel and cobalt totals for 2035 with plausible remainders (International Energy Agency, 2026b).

A2.3 Backtest

The backtest fits a log-linear trend to each series through 2019. Table A2 compares the resulting projections with actual outcomes from 2020 to the latest year available. Only series adjusted for inflation change the growth rate. Nominal OECD STAN output series only widen the ranges, because inflation after 2021 would otherwise appear as growth in physical output.

Table A2: Backtest of tier 1 drivers, trend to 2019 against outcome

Series	Industry	Basis	Test years	MAPE %	Bias %	Growth error, pp a year	Corrects growth
IEA upstream investment	Oil and gas	Real	2020 to 2025	17.3	−7.6	−3.45	Yes
IEA gas power investment	Energy	Real	2020 to 2025	36.5	−36.0	−20.68	Yes
IEA nuclear investment	Energy	Real	2020 to 2025	50.3	−50.3	−16.42	Yes
OECD hip and knee procedures	Medical	Counts	2020 to 2024	9.7	9.2	−0.53	Yes
OECD STAN C30 output	Aerospace	Nominal	2020 to 2023	14.9	14.9	0.31	No
OECD STAN C26 output	Electronics	Nominal	2020 to 2023	7.5	−6.2	−2.58	No
OECD STAN C27 output	Electronics	Nominal	2020 to 2023	12.8	−11.0	−5.65	No
OECD STAN C20 output	Chemical	Nominal	2020 to 2023	13.8	−9.9	−4.31	No
OECD STAN C29 output	Automotive	Nominal	2020 to 2023	14.8	14.8	−1.07	No
Eurostat C29 volume, EU27	Automotive	Real	2020 to 2025	23.0	23.0	3.22	Yes
Eurostat C29 volume, DE, ES, FR, IT	Automotive	Real	2020 to 2025	17.2 to 49.1	17.2 to 49.1	2.02 to 7.02	No
IEA WEI 2025 against WEI 2026	Oil and gas	Real	2025	5.2	5.2	n/a	No

Sources: team calculation (Product Strategy 2035 Team, 2026a) on IEA, OECD and Eurostat data (Eurostat, 2026b; International Energy Agency, 2026j; OECD, 2025, 2026). Bias is the mean of projection minus outcome over outcome. The growth error is the trend slope minus the realised slope.

The backtest raised the annual growth rate by 1.0 percentage point for energy and for oil and gas, and by 0.27 points for medical technology. It lowered the rate by 1.0 point for automotive. The rates for aerospace, chemicals, and electronics stayed the same.

Without these adjustments, the P50 indices would be 142 for energy, 104 for oil and gas, 137 for medical technology, and 108 for automotive. The earlier fixed-rate workbook gave 148, 105, 128, and 86 for those sectors, respectively. The new model replaces those earlier workbook results (Product Strategy 2035 Team, 2026a).

A2.4 Further tier 2 candidates

Table 15 in chapter 7 lists eight opportunities. Table A3 adds the remaining register entries and candidates from the team's research workbook that have yet to be assessed. These opportunities remain outside tier 1.

Table A3: Further tier 2 candidates

Opportunity	Alloy and product form	TRL and evidence	Earliest volume	Signpost or status
Molten salt reactor vessels and piping	NiMoCr plate and tube	3 to 5, speculative	2035 or later	Construction permit and ASME code case
Superhot and enhanced geothermal casing	Nickel alloy tube and clad pipe	4 to 6, speculative	2032 to 2035	Well above 400 °C with a published casing alloy
Liquid hydrogen aircraft tanks	Aluminium sheet, austenitic tube	3 to 4, speculative	2038 or later	Airframer programme launch
Concentrating solar receivers, Gen 3	Nickel alloy plate, tube, forgings	Not assessed	Not assessed	Workbook candidate
Fuel cell bipolar plates, stationary PEM	Coated stainless thin strip	Not assessed	Not assessed	Workbook candidate
Robotic surgery components	NiTi wire and precision tube	Not assessed	Not assessed	Workbook candidate
Smaller cardiovascular delivery systems	CoNiCrMo fine wire and tube	Not assessed	Not assessed	Workbook candidate
Compact electronics cooling	CuCrZr and copper powder	Not assessed	Not assessed	Workbook candidate

Sources: team register and team research workbook, TRL and years are team estimates (Product Strategy 2035 Team, 2026a, 2026b).

A2.5 Scoring rules for the opportunity matrix

Table 16 scores opportunities for a qualified producer that already melts and rolls high-performance alloys. Table A4 explains how to assign scores from 1 to 5. Scores of 2 and 4 fall between the listed anchor points. For early-stage opportunities, the growth score uses the earliest year of commercial volume because the model cannot quantify growth. To calculate the total, multiply each score by its weight and add the results.

Table A4: Scoring rules, weights and evidence

Criterion, weight	Evidence used	Score 5	Score 3	Score 1
Market attractiveness, 15%	Measured 2025 pool and unit value (chapters 5 and 6)	Largest measured pools, such as aviation cobalt or EU flat products of EUR 1.8 to 2.8 bn	Measured niche with premium unit value	No measured volume yet
Growth to 2035, 20%	P50 growth a year of the most specific index in Tables 13 and 14 or driver in Table A1, tier 2 by earliest volume	4.0% a year or more (4 from 3.0%)	2.0 to 2.9% a year (2 from 0%), or tier 2 volume by 2028	Below 0%, or tier 2 volume after 2033
Supply security, 15%	Concentration, quotas and licences of the main inputs (chapter 4, supply overlay)	Nickel, iron or chromium inputs without export restriction	Cobalt as a minor element under the Congo quota	An input under an active licence halt
Competitive intensity, 10%	Number of qualified suppliers and long term agreements (A6)	Demand above qualified capacity, no dominant incumbent	Several qualified suppliers competing on service	Price competition with new Asian capacity (2 for an oligopoly with agreements)
Technology attractiveness, 10%	Process route and how many producers master it	Triple melt, foil below 0.1 mm or similar rare routes	Standard grade on a qualified route	Commodity grade in decline
Customer attractiveness, 10%	Buyer backlog, concentration and pricing (chapter 5)	Named buyers with sold out capacity or growing backlog	Fragmented or cyclical buyers	Collapsing orders or state price cuts
Entry barriers, 10%	Time to approval and protection afterwards (A7)	Passable within about two years and protective thereafter (4 for three to five years)	Programme specific or moderate approvals	Captive supply chain closed to outsiders
Differentiation potential, 10%	Scope for grade, tolerance or service advantage	Customer specific grades or tolerances	Partly standardised products	Interchangeable products

Source: team scoring rules (Product Strategy 2035 Team, 2026a). Priority 1 from a weighted score of 3.75, priority 2 from 3.00, and early signal rows are at most gated options. Under equal weights the three priority 1 rows score 4.00, 4.00 and 3.75, and with entry barriers at 20 percent they score 4.14, 4.00 and 3.82.

A3 Data sources, product codes and reproducibility

The team collected structured data with the project's fetch script. The script saves raw files, creates clean tables, and records each query's date and URL in a source register with more than 1,000 entries.

The UN Comtrade data cover HS 7502, 7504, 7505, 7506, 7507, 8105, 8109, 7219, 7220, 7223, 7409, 7606, 8311, and 8411 for ten reporting countries from 2019 to 2024. Eurostat PRODCOM covers fifteen nickel, cobalt, stainless, copper, and aluminium product codes for the EU and five producer countries. Other sources include USGS Mineral Commodity

Summaries 2026, the World Bank Pink Sheet, OECD STAN and ICIO, UNIDO INDSTAT for Germany, and filings from 61 companies.

Table 10 in chapter 6 estimates market values by multiplying EU apparent consumption by the lower and higher of the 2024 unit values for exports and imports outside the EU. The team used these figures because Eurostat production values fall below the value of the nickel contained in the products.

For flat products, 57.8 kt × EUR 31.0 per kg gives EUR 1.79 bn, and 57.8 kt × EUR 47.8 per kg gives EUR 2.76 bn. For tubes, 9.2 kt × EUR 26.1 per kg gives EUR 0.24 bn, and 9.2 kt × EUR 41.1 per kg gives EUR 0.38 bn. The team did not value bars because the production figure is confidential (Eurostat, 2025c, 2026d).

The team calculated unit values by dividing trade value by trade quantity in PRODCOM DS-059358. The resulting range reflects the mix of grades traded across the EU border. It does not represent a quoted list price.

A4 Limitations and further research

The study has three main limitations. First, official data stop at the mill product stage and combine pure nickel with nickel alloys. This makes it hard to estimate the market for specific alloys. A survey of producers or customs data at the eight-digit level could improve the estimates.

Second, tier 1 uses published activity drivers, but its alloy intensity, use shares, and product family weights are team assumptions. Nickel alloy tonnes by industry are not published, and the backtest periods are short. The team should rerun the model when the IEA World Energy Outlook 2026, the next Airbus and Boeing outlooks, and new SEMI forecasts become available.

Third, the trade analysis covers ten large reporting countries. A global analysis of mirror data that includes Russia, Sweden, and Austria would give a fuller picture of trade routes.

A5 Sector market indicators

The following tables list the market indicators used in chapter 5. Each table identifies the sources and years of observation.

Table A5: Automotive market indicators, 2025 and 2035

Indicator	2025	2035	Source basis
World motor vehicle production	96.4 million	No official forecast	OICA

Indicator	2025	2035	Source basis
Electric share of new car sales, world	25%	About 50%	IEA, both exploratory scenarios
Electric share of car sales, China	Close to 55%	Above 90%	IEA
Electric share of car registrations, EU	26.8% (BEV 17.4% plus PHEV 9.4%, team sum)	Above 90% of light duty sales	ACEA, IEA Stated Policies
Turbocharger production	Close to 50 million units	Back toward 2022 level by 2030	Garrett citing S&P Global Mobility

Sources: OICA (2026), ACEA (2026), IEA Global EV Outlook 2026, Garrett Motion 10-K 2025 (European Automobile Manufacturers' Association, 2026a, 2026b; Garrett Motion, 2026; International Energy Agency, 2026c; International Organization of Motor Vehicle Manufacturers, 2026).

Table A6: Oil and gas indicators, today and 2035

Indicator	Today	2035	Source basis
Upstream oil and gas investment	USD 570 bn (2025), USD 546 bn (2026)	Near today's level (Stated Policies), about 10% higher (Current Policies)	IEA
Oil demand	100 mb/d (2024)	Peak near 102 mb/d around 2030, back near 100 mb/d	IEA Stated Policies
LNG trade	560 bcm (2024)	880 bcm (Current Policies)	IEA
Upstream growth regions 2026	Africa +12%, Central and South America +10%	Continued shift	IEA

Sources: IEA World Energy Outlook 2025 and World Energy Investment 2026 (International Energy Agency, 2025c, 2026h).

Table A7: Energy technology indicators relevant to high performance metals

Indicator	Latest value	Direction to 2035	Metal and product form
Siemens Energy gas turbine capacity	About 17 GW (FY2024), sold out to FY2028	Above 30 GW a year from FY2028	Superalloy bar, billet and forging stock
Reactors under construction	81 reactors, 88.5 GWe gross, 40 in China	710 GW installed in 2035 (IEA Net Zero)	Zirconium cladding tube, Alloy 690 tube
Installed electrolysis	More than 4 GW (2025)	Committed 2030 output 4.3 Mt of hydrogen	Nickel strip and mesh
Solid oxide fuel cells, Bloom Energy	1 GW a year, doubling to 2 GW by end 2026	Driven by data centre power	Interconnect material not disclosed

Sources: Siemens Energy capital markets day 2025, World Nuclear Association (2026), IEA World Energy Outlook 2025 and Global Hydrogen Review 2026, Bloom Energy results (Bloom Energy, 2026; International Energy Agency, 2025c, 2026e; Siemens Energy, 2025; World Nuclear Association, 2026b).

Table A8: Aerospace market indicators, today and 2035

Indicator	Today	2035	Source basis
Passenger fleet, aircraft of 100 seats or more	23,310 (2025)	About 32,600 (team interpolation)	Airbus GMF 2026
Commercial fleet, wider definition	About 28,000 (2025)	About 38,000	Boeing CMO 2026
New generation share of fleet	39% (2025)	82%	Airbus
Aviation services market	USD 209 bn (2026)	USD 422 bn in 2045, 3.8% a year	Airbus GSF 2026
Cobalt used in aviation	11,499 t (2025)	Grows with fleet and engine output	Cobalt Institute
World military expenditure	USD 2,887 bn (2025), Europe +14%	Rising in Europe	SIPRI

Sources: Airbus (2026), Boeing (2026), Cobalt Institute (2026), SIPRI (2026) (Airbus, 2026a, 2026b; Boeing, 2026; Cobalt Institute, 2026; Stockholm International Peace Research

Institute, 2026). The 2035 passenger fleet is a team estimate on a constant growth path and not an Airbus figure.

Table A9: Chemical industry indicators

Indicator	Value	Implication for alloy demand
EU27 chemical sales, 2024	About EUR 635 bn	Large installed base needing maintenance
EU share of world chemical sales	13% (2024), from 21% (2009)	New build moves out of Europe
EU chemical production index, 2025	Down 3.2%	Fewer debottlenecking projects
EU capacity utilisation, Q3 2026	75.0% against 81.3% average	Closures, not expansions
German utilisation, 2025	72.5%	Weakest home market
Asia Pacific share of growth to 2035	Over 80% (BASF)	Export projects through fabricators

Sources: Cefic (2025, 2026), Eurostat, VCI, BASF Report 2025 (BASF, 2026; European Chemical Industry Council, 2025a, 2025b, 2026b; Eurostat, 2026c; Verband der Chemischen Industrie, 2026).

Table A10: Electronics indicators relevant to high performance metals

Indicator	Value	Relevance
World semiconductor sales	USD 795.6 bn (2025), USD 1.51 tn (2026 forecast)	Activity, not metal volume
300 mm fab equipment spending	USD 149 bn (2026), USD 227 bn (2029)	Gas delivery and chamber parts in high purity alloys
Copper busbar in a 1 MW rack at 54 VDC	Up to 200 kg	Copper and copper alloy strip and bar
German electrical and electronics value added	EUR 112.2 bn (2023), real output flat in 2025	Home market for sensor and seal alloys
DNP metal mask capacity	Doubled by the 8th generation Kurosaki line	FeNi36 (invar) ultra thin foil

Sources: WSTS (2026), SEMI (2026), NVIDIA (2025), OECD STAN and ZVEI, Dai Nippon Printing (Dai Nippon Printing, 2024a; NVIDIA, 2025a; OECD, 2026; SEMI, 2026a; World Semiconductor Trade Statistics, 2026; ZVEI, 2026).

Table A11: Medical technology indicators

Indicator	Value	Relevance
Hip replacements per 100,000 people, OECD average	198 (2023)	Volume driver for CoCr and zirconium bar
Knee replacements per 100,000 people, OECD average	156 (2023)	Volume driver for CoCr bar and powder
European medtech market	About EUR 173 bn (2025)	Devices and diagnostics, not metal
Stryker orthopaedics revenue	USD 9,469 m (2025)	Largest implant buyer
Zimmer Biomet knees and hips	USD 3,322 m and USD 2,094 m (2025)	Names cobalt chrome as a purchased commodity
China volume based procurement	Implant prices cut by about 82% (2022)	Volume without margin

Sources: OECD Health at a Glance 2025, MedTech Europe, company filings, National Healthcare Security Administration (MedTech Europe, 2026; National Healthcare Security Administration, 2022; OECD, 2025; Stryker, 2026; Zimmer Biomet, 2026).

A6 Peer benchmark of specialty metal producers

The peer comparison supports the assessment of competition in section 8.1. It shows the margins of leading specialty metal producers and how they are changing capacity.

Table A12: Peer benchmark of specialty metal producers

Company	Revenue, latest year	Margin	Capacity move
Howmet Aerospace	USD 8,252 m (2025)	24.8% operating	Engine products capacity, USD 453 m capex
Carpenter Technology	USD 3,124 m (FY2026)	22.5% operating	Athens, Alabama expansion, capex USD 355 to 375 m guided
ATI	USD 4,587 m (2025)	14.0% operating	Fifth VIM furnace in Monroe, about 9 kt a year from 2027
PCC including Special Metals	USD 10.8 bn (2025)	Not published	Aerospace revenue up 7.5%
Aubert & Duval	EUR 844 m (2024)	Not published	Owned by Airbus, Safran and Tikehau
Alleima	SEK 18,630 m (2025)	8.3% adjusted EBIT	Fourth VAR furnace at Sandviken

Company	Revenue, latest year	Margin	Capacity move
Outokumpu	EUR 5,468 m (2025)	3.1% adjusted EBITDA	About 13 kt high nickel flat at Avesta, about EUR 150 m
Aperam	EUR 6,080 m (2025)	5.6% adjusted EBITDA	New VIM complex at Imphy within EUR 160 m programme
Daido Steel	JPY 578 bn (FY2025)	7.3% operating	JPY 36 bn high alloy melting and forging base
Nippon Yakin Kogyo	JPY 151 bn (FY2025)	7.3% operating	Sales down 12% on weak high nickel demand

Sources: company filings and reports (Alleima, 2026a; Aperam, 2026; Argus Media, 2026b; Armees.com, 2025; ATI, 2026a; Berkshire Hathaway, 2026; Carpenter Technology, 2026; Daido Steel, 2026; Howmet Aerospace, 2026; Nippon Yakin Kogyo, 2026; Outokumpu, 2026a, 2026b). Margins as reported or calculated from reported figures, fiscal years differ.

A7 Qualification gates by industry

The qualification requirements below affect how quickly a producer can enter each application. They support the entry barrier scores in Table 16 and the conditions for proceeding in Table 17.

Table A13: Qualification gates by industry

Industry	Approval gate	Who approves	Implication
Aerospace	AMS specifications, NADCAP accredited processes, part level approval	Engine makers and forgers	Years per part family, start in 2026
Energy	Casting house and turbine maker approvals, ASME III and RCC-M for nuclear	Turbine makers, utilities, tube makers	Nuclear approvals only with a named programme
Oil and gas	NORSOK M-650, API 6ACRA and API 5LD, operator approval lists	Operators, pipe and equipment makers	Approval lists are the moat
Chemical	Pressure equipment rules (VdTÜV, AD 2000, ASME), corrosion test data	Fabricators, plant operators	Life cycle cost evidence wins
Medical	ASTM F75 and F1537 material standards, supplier audits	Implant makers with ISO 13485	Few buyers, long relationships

Source: team summary of the standards named in chapter 5.

A8 Risks and kill criteria

The roadmap in Table 17 identifies risks. Table A14 gives an early signal, a response, and a point at which a producer should stop for each risk.

Table A14: Main risks, responses and kill criteria

Risk	Early signal	Response	Kill criterion
Qualification slower than planned	Part approvals behind schedule	Focus on repair and casting stock, which qualify faster	No long term agreement by end 2028, so no capacity step
Substitution by ceramic matrix composites and titanium aluminide	New engine variants with lower nickel content	Join customer development programmes early	Falling hot section nickel content in new programmes
New Asian superalloy capacity	Japanese and Chinese producers win Western approvals	Speed and service close to European customers	Prices below conversion cost in tenders
Cobalt, tungsten and molybdenum supply	Quota and licence changes	Several sources, revert recovery, stocks	Single source above half of an input
European demand weakness	Chemical utilisation below 75%, falling engine volumes	Shift sales to Asia and the United States	Contribution margin below plan for two years

Source: team assessment based on chapters 4 to 7 (European Chemical Industry Council, 2026b; SteelOrbis, 2026).